Stochastic Calculus and Option Pricing
Synthesis Exercise Sheet

1 Week 1 — Option Pricing and Core Probability Tools

Focus. Random variables, distributions, and characteristic functions; sample statistics, the Law of Large Numbers, and the Central Limit Theorem; joint distributions and information; conditional expectation; processes, filtrations, and martingales; and a short differential-calculus review. Exercises marked "Extension" may be left for additional practice.

Part I — Random Variables and Distributions

Exercise 1.1 (Level 1 — A financial payoff as a random variable)**.** At maturity $T = 1$, a stock price has the physical distribution

s	70	105	140
$\mathbb{P}(S_T = s)$	0.25	0.50	0.25

and $X = (S_T - 100)^+$ is the payoff of a European call.

(a) Identify the underlying, maturity, strike, and possible values of X.

(b) Determine the law and the cdf of X.

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$.

(d) If $r = 3\%$ is continuously compounded, compute $e^{-rT}\mathbb{E}[X]$. Explain why this discounted physical expectation is not, by itself, a no-arbitrage price.

Exercise 1.2 (Level 1 — Cdf, density, and moments)**.** Let $U \sim \mathrm{Unif}[-1, 3]$.

(a) Write the cdf F_U on the whole real line and recover the density f_U.

(b) Compute $\mathbb{P}(0 < U \leq 2)$ from the cdf and again from the density.

(c) Starting from the defining integrals, compute $\mathbb{E}[U]$, $\mathbb{E}[U^2]$, and $\mathrm{Var}(U)$.

(d) Set $V = 2U + 1$. Find its support, cdf, mean, and variance.

Exercise 1.3 (Level 2 — Empirical moments, LLN, and CLT)**.** The observed daily claim counts in a short sample are

$$2,\ 5,\ 4,\ 3,\ 6.$$

In the population, suppose daily claim counts $X_1, X_2, \ldots$ are i.i.d. Poisson with parameter 4.

(a) Compute the sample mean, the empirical variance with denominator n, and the unbiased sample variance with denominator $n - 1$.

(b) State the LLN for $\overline{X}_n$ and interpret it here.

(c) State the CLT for $\overline{X}_n$. What are the approximate mean and standard deviation of $\overline{X}_{100}$?

(d) Use the CLT to approximate $\mathbb{P}(\overline{X}_{100} > 4.3)$. You may use $\Phi(1.5) \approx 0.9332$.

(e) Explain in one sentence the different conclusions supplied by the LLN and the CLT.

Exercise 1.4 (Level 2 — Characteristic function of a Gaussian)**.** Let $Z \sim \mathcal{N}(0,1)$ and define

$$\varphi_Z(u) = \mathbb{E}[e^{iuZ}], \qquad u \in \mathbb{R}.$$

The following steps derive the characteristic function without using a complex change of variables.

(a) Using the standard Gaussian density, write $\varphi_Z(u)$ as an integral and verify that $\varphi_Z(0) = 1$.

(b) Differentiate under the integral sign to express $\varphi_Z'(u)$. You may use

$$\frac{\mathrm{d}}{\mathrm{d}z} e^{-z^2/2} = -z e^{-z^2/2}.$$

(c) Integrate by parts and show that

$$\varphi_Z'(u) = -u\varphi_Z(u).$$

Check explicitly that the boundary term vanishes.

(d) Solve this differential equation using the initial condition from part (a), and obtain $\varphi_Z(u)$.

(e) If $X = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$, deduce that

$$\varphi_X(u) = \exp\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right).$$

(f) Let X_1 and X_2 be independent Gaussian variables. Use characteristic functions to identify the distribution of $X_1 + X_2$.

Part II — Joint Distributions and Information

Exercise 1.5 (Level 1 — Reading a joint law)**.** The joint law of (X, Y) is

	$Y = L$	$Y = H$	$\mathbb{P}(X = x)$
$X = 0$	0.10	0.40	
$X = 1$	0.20	0.30	
$\mathbb{P}(Y = y)$			1

(a) Complete the marginal probabilities in the table.

(b) Are X and Y independent? Give a numerical justification.

(c) Compute $\mathbb{P}(X = 1 \mid Y = L)$ and $\mathbb{P}(X = 1 \mid Y = H)$.

(d) Interpret X as a loss indicator and Y as a rating signal. Which signal carries the larger conditional loss probability?

For a signal Y, $\sigma(Y)$ denotes exactly the information revealed by observing Y. On a finite space, its events are the unions of the level sets of Y.

Exercise 1.6 (Level 2 — Information generated by a signal)**.** A fair die is rolled and $Y = \mathbf{1}_{\{4,5,6\}}$. Let $\mathcal{G} = \sigma(Y)$.

(a) Give the partition of $\Omega = \{1, \dots, 6\}$ generated by Y and list all events in $\mathcal{G}$.

(b) Decide whether each variable is $\mathcal{G}$-measurable:

$$Z_1 = Y, \qquad Z_2 = (-1)^Y, \qquad Z_3 = \mathbf{1}_{\{5,6\}}, \qquad Z_4 = 2 + 3Y.$$

Justify each answer using constancy on the atoms of the partition.

(c) Find $g : \{0, 1\} \to \mathbb{R}$ such that $Z_2 = g(Y)$.

(d) Explain what information is missing when one observes Y but wants to determine Z_3.

Part III — Conditional Expectation

Exercise 1.7 (Level 1 — Conditional expectation on a finite partition)**.** A fair die is rolled. The available information reveals only the block

$$L = \{1, 2\}, \qquad M = \{3, 4\}, \qquad H = \{5, 6\}, \qquad \mathcal{G} = \sigma(L, M, H).$$

Define $X = 6\mathbf{1}_{\{\omega \geq 4\}}$.

(a) Is X $\mathcal{G}$-measurable? Explain.

(b) Compute $\mathbb{E}[X]$ and the average of X on each atom L, M, H.

(c) Write $\mathbb{E}[X \mid \mathcal{G}]$ using indicator functions.

(d) Verify the defining averaging property on L, M, and H.

(e) Verify $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.

Exercise 1.8 (Level 2 — Conditional density and conditional moments)**.** The pair (X, Y) has joint density

$$f_{X,Y}(x, y) = (x + y)\mathbf{1}_{\{0<x<1,\ 0<y<1\}}.$$

(a) Verify that the density integrates to one and compute both marginal densities.

(b) For $0 < y < 1$, derive $f_{X|Y}(x \mid y)$ and its support. Are X and Y independent?

(c) Compute $\mathbb{E}[X \mid Y = y]$, then write $\mathbb{E}[X \mid Y]$ as a random variable.

(d) For $f(x) = x^2$, compute $\mathbb{E}[f(X) \mid Y = y]$. Identify g such that $\mathbb{E}[f(X) \mid Y] = g(Y)$ and explain why this is $\sigma(Y)$-measurable.

(e) Verify the tower property by comparing $\mathbb{E}[\mathbb{E}[X \mid Y]]$ with $\mathbb{E}[X]$.

Exercise 1.9 (Extension — Gaussian conditioning)**.** Let (X, Y) be jointly Gaussian with

$$\mathbb{E}[X] = \mu_X, \quad \mathbb{E}[Y] = \mu_Y, \quad \mathrm{Var}(X) = \sigma_X^2, \quad \mathrm{Var}(Y) = \sigma_Y^2 > 0,$$

and $\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.

(a) Set $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ and $R = X - \mu_X - a(Y - \mu_Y)$. Show that $\mathrm{Cov}(R, Y) = 0$.

(b) Use joint Gaussianity to conclude that R and Y are independent.

(c) Deduce formulas for $\mathbb{E}[X \mid Y]$ and $\mathrm{Var}(X \mid Y)$.

(d) Interpret $|\rho|$ as hedge effectiveness and the conditional variance as residual risk.

Part IV — Stochastic Processes and Filtrations

Exercise 1.10 (Level 2 — Three coin tosses viewed through time)**.** Three fair coins are tossed. Write $\omega = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{H, T\}^3$ and define

$$\mathcal{F}_0 = \{\varnothing, \Omega\}, \qquad \mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n), \quad n = 1, 2, 3.$$

Let S_n be the number of heads observed by time n.

(a) Describe the atoms of $\mathcal{F}_0, \ldots, \mathcal{F}_3$ and verify that these sigma-algebras form a filtration.

(b) Show that (S_0, S_1, S_2, S_3) is adapted.

(c) Define $A_0 = A_1 = 0$, $A_2 = \mathbf{1}_{\{\varepsilon_3 = H\}}$, and $A_3 = A_2$. At which time does (A_n) fail to be adapted? Interpret the failure.

(d) Let τ be the first head, capped at 3. Determine τ on every outcome and verify $\{\tau \le n\} \in \mathcal{F}_n$.

Part V — Martingales and Fair Games

Exercise 1.11 (Level 2 — A fair but asymmetric random walk)**.** Let $(\xi_k)_{k \ge 1}$ be independent with

$$\mathbb{P}(\xi_k = 2) = \frac{1}{3}, \qquad \mathbb{P}(\xi_k = -1) = \frac{2}{3}, \qquad S_n = \sum_{k=1}^{n} \xi_k, \qquad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n).$$

(a) Compute $\mathbb{E}[\xi_k]$ and $\mathbb{E}[\xi_k^2]$. Why is this a fair game?

(b) Prove from the definition that (S_n) is a martingale.

(c) Prove that $S_n^2 - 2n$ is a martingale.

(d) If instead $\mathbb{P}(\xi_k = 2) = 1/4$, find the predictable drift and construct a centered martingale from S_n.

(e) Explain the casino interpretation of the martingale property.

Exercise 1.12 (Level 2 — Replication and risk-neutral pricing)**.** A stock has $S_0 = 50$ and, after one period,

$$S_1 = 65 \quad \text{or} \quad S_1 = 45.$$

The risk-free gross return is $R = 1.05$. A call has strike $K = 55$ and payoff $H = (S_1 - K)^+$.

(a) Build the state-by-state payoff table.

(b) Find the risk-neutral up probability q from $S_0 = R^{-1}\mathbb{E}^q[S_1]$ and check that $q \in (0, 1)$.

(c) Compute $V_0 = R^{-1}\mathbb{E}^q[H]$.

(d) Find Δ shares and a maturity cash position B_1 such that $\Delta S_1 + B_1 = H$ in both states.

(e) Check that $\Delta S_0 + B_1/R = V_0$ and explain why the physical up probability does not determine this price.

(f) Relate the pricing identity to the martingale property of discounted asset prices under the risk-neutral probability.

Part VI — Mathematical Toolbox

Exercise 1.13 (Levels 1–2 — Derivatives, gradients, and the chain rule). **(a)** Starting from the difference quotient, compute the derivative of $g(x) = x^2$.

(b) For $f(x, y) = x^2 y + e^y$, compute all first and second partial derivatives, the gradient, and the Hessian.

(c) Let S be differentiable and let $V = V(t, s)$ be smooth. If $h(t) = V(t, S(t))$, derive $h'(t)$ from the multivariable chain rule.

(d) Apply the previous result to $V(t, s) = e^{-rt} s^2$.

(e) The payoff $v(s) = (s - K)^+$ is continuous but not differentiable at one point. Identify it and compute the derivative everywhere else.

2 Week 2 — Brownian Motion, Stopping Times, and Quadratic Variation

Focus. Gaussian structure and covariance kernels; Brownian transformations; Markov renewal and conditional expectation; stopping times and strong Markov property; reflection and hitting times; quadratic variation; geometric Brownian motion; and adapted trading gains. All calculations are kept symbolic.

Part I — Gaussian Structure and Brownian Transformations

Exercise 2.1 (Level 2 — Gaussian vectors and the covariance kernel). Let $(W_t)_{t \geq 0}$ be a standard Brownian motion and fix $0 < r < s < t$.

(a) Find the joint law of the increment vector

$$(W_r,\, W_s - W_r,\, W_t - W_s).$$

(b) Deduce that (W_r, W_s, W_t) is Gaussian and compute its covariance matrix.

(c) Compute the conditional distribution of W_t given $\mathcal{F}_s$, then given $W_s = x$.

(d) Show that the Brownian covariance kernel $K(u, v) = \min(u, v)$ is positive semidefinite, meaning that

$$\sum_{i,j=1}^{n} a_i a_j K(t_i, t_j) \geq 0$$

for all dates $t_1, \ldots, t_n$ and coefficients $a_1, \ldots, a_n$.

(e) Explain why the mean function and the covariance kernel determine every finite-dimensional Brownian law.

Exercise 2.2 (Level 2 — Symmetry, scaling, renewal, and time reversal). Fix $c > 0$ and $s > 0$. Consider

$$X_t = -W_t, \qquad Y_t = \frac{W_{ct}}{\sqrt{c}}, \qquad Z_t = W_{s+t} - W_s, qquadt \geq 0,$$

and $R_t = W_{s-t} - W_s$ for $0 \leq t \leq s$.

(a) Verify from the defining properties that X and Y are standard Brownian motions.

(b) Show that Z is a standard Brownian motion independent of $\mathcal{F}_s$. Interpret this as deterministic-time renewal.

(c) Show that R has the same finite-dimensional laws as Brownian motion restricted to $[0, s]$.

(d) Deduce the square-root-of-time identity

$$\max_{0 \le u \le ct} W_u \stackrel{d}{=} \sqrt{c} \max_{0 \le u \le t} W_u.$$

(e) State the almost-sure lim sup and lim inf identities from the law of the iterated logarithm, both as $t \to \infty$ and as $t \downarrow 0$.

Part II — Markov Property, Martingales, and Stopping

Exercise 2.3 (Level 2 — Markov semigroup and martingale correction)**.** For bounded measurable f and $h > 0$, define

$$(P_h f)(x) = \int_{\mathbb{R}} f(x+y) \frac{e^{-y^2/(2h)}}{\sqrt{2\pi h}} \, \mathrm{d}y.$$

(a) Prove the conditional-expectation form of the Markov property:

$$\mathbb{E}[f(W_{t+h}) \mid \mathcal{F}_t] = (P_h f)(W_t).$$

(b) Prove the semigroup identity $P_{h+k} = P_h P_k$.

(c) For $0 \le s < t$, compute $\mathbb{E}[W_t \mid \mathcal{F}_s]$ and $\mathbb{E}[W_t^2 \mid \mathcal{F}_s]$.

(d) Deduce directly that W_t and $W_t^2 - t$ are martingales.

(e) State the general proposition defining $\langle X \rangle$ for a continuous square-integrable martingale X, and identify $\langle W \rangle_t$.

Exercise 2.4 (Level 2 — Stopping times and information at stopping)**.** Fix $T > 0$, $a > 0$, and $\ell < 0 < u$. Consider

$$\tau_a = \inf\{t \ge 0 : W_t = a\}, \qquad \tau_{\ell,u} = \inf\{t \ge 0 : W_t \notin (\ell, u)\},$$

$$\rho_T = \sup\{0 \le t \le T : W_t = 0\}, \qquad \gamma_T = \arg\max_{0 \le t \le T} W_t.$$

(a) Determine which of $T, \tau_a, \tau_{\ell,u}, \rho_T, \gamma_T$ are stopping times and justify each answer.

(b) Prove, using path continuity, that

$$\{\tau_a \le t\} = \left\{ \max_{0 \le v \le t} W_v \ge a \right\}.$$

(c) Give the definition of $\mathcal{F}_\tau$. Explain why W_τ is $\mathcal{F}_\tau$-measurable for an almost surely finite stopping time τ.

(d) State the strong Markov property and apply it at τ_a.

(e) Explain precisely why strong Markov, rather than only deterministic Markov, is needed for the reflection principle.

Part III — Reflection and Hitting Times

Exercise 2.5 (Level 2 — Reflection principle and first passage)**.** Let

$$M_t = \max_{0 \le u \le t} W_u, \qquad \tau_a = \inf\{u \ge 0 : W_u = a\}, \qquad a > 0.$$

(a) Define the reflected path after τ_a and describe the one-to-one correspondence between the relevant terminal events.

(b) Derive the distribution functions of M_t and τ_a.

(c) Differentiate to obtain the density of τ_a.

(d) Show that $M_t \overset{d}{=} |W_t|$, then compute $\mathbb{E}[M_t]$.

(e) Use Brownian scaling to prove

$$\tau_a \overset{d}{=} a^2 \tau_1.$$

(f) Show that $\mathbb{P}(\tau_a < \infty) = 1$, but $\mathbb{E}[\tau_a] = \infty$.

Part IV — Quadratic Variation

Exercise 2.6 (Level 2 — Realized quadratic variation)**.** Fix $T > 0$, let $t_i = iT/n$, and define

$$Q_n = \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2.$$

(a) Compute $\mathbb{E}[Q_n]$ and $\mathrm{Var}(Q_n)$.

(b) Deduce that $Q_n \to T$ in L^2.

(c) Let A be continuously differentiable. Prove that $[A]_T = 0$.

(d) Show that the cross-term

$$\sum_i (A_{t_{i+1}} - A_{t_i})(W_{t_{i+1}} - W_{t_i})$$

converges to zero in probability.

(e) Deduce that, for $X_t = A_t + \sigma W_t$, $[X]_T = \sigma^2 T$.

(f) If X is observed on the grid, propose a consistent estimator of σ^2 based on its squared increments.

Part V — Price Models and Trading Gains

Exercise 2.7 (Level 2 — Symbolic analysis of geometric Brownian motion)**.** Let $S_0 > 0$, $\sigma > 0$, and

$$S_t = S_0 \exp\left(\left(\mu - \tfrac{1}{2}\sigma^2\right) t + \sigma W_t\right).$$

(a) Find the law of $\log(S_t/S_0)$ and the conditional law of $\log(S_t/S_s)$ given $\mathcal{F}_s$, for $0 \le s < t$.

(b) Compute $\mathbb{E}[S_t]$, $\mathrm{Var}(S_t)$, and the median of S_t.

(c) For a level $K > 0$, express $\mathbb{P}(S_t > K)$ using Φ.

(d) Compute $[\log S]_t$ and explain why the drift does not appear.

(e) For a partition of $[0, T]$, show how squared log-returns provide a consistent estimator of σ^2 as the mesh tends to zero.

Exercise 2.8 (Level 2 — A two-step adapted trading gain)**.** Fix $0 < t_1 < T$. Let $\Delta_0 \in \mathbb{R}$ and let Δ_1 be a square-integrable $\mathcal{F}_{t_1}$-measurable random variable. Define

$$G_T = \Delta_0 W_{t_1} + \Delta_1 (W_T - W_{t_1}).$$

(a) Explain why the measurability of Δ_1 is the no-look-ahead condition.

(b) Prove that $\mathbb{E}[G_T] = 0$.

(c) Show that the two gain terms are orthogonal in L^2, and deduce

$$\mathbb{E}[G_T^2] = \Delta_0^2 t_1 + (T - t_1)\mathbb{E}[\Delta_1^2].$$

(d) Explain why choosing $\Delta_1 = \mathbf{1}_{\{W_T - W_{t_1} > 0\}}$ is inadmissible.

(e) Write the corresponding piecewise constant process Δ_t and express G_T as $\int_0^T \Delta_t \, \mathrm{d}W_t$.

3 Week 3 — Stochastic Integration: Construction and Applications

Focus. Left-point gains and predictability; simple stochastic integrals; the Itô isometry and the L^2 extension; the integrability required for true martingales; conditional variance, stopping, and quadratic variation; deterministic Gaussian integrals; deterministic integration by parts; left/right/midpoint conventions; and simulation diagnostics. No general Itô formula is used.

Part I — Predictability and Simple Integrals

Exercise 3.1 (Level 1 — Information available at the trading time)**.** Let $(W_t)_{t\geq 0}$ be Brownian motion with its natural filtration. On the grid $0, 1, 2$, write $\Delta W_i = W_{i+1} - W_i$.

(a) For each rule below, determine whether the position used on $(i, i+1]$ is predictable:

(i) $\phi_0 = 1$, $\phi_1 = 1 + W_1^2$;

(ii) $\phi_0 = \operatorname{sign}(W_1)$, $\phi_1 = 0$;

(iii) $\phi_0 = 0$, $\phi_1 = \mathbf{1}_{\{W_1 > 0\}}$;

(iv) $\phi_i = \operatorname{sign}(\Delta W_i)$.

Justify each answer with the relevant measurability statement.

(b) For every admissible rule, write the terminal gain explicitly as a weighted sum of Brownian increments.

(c) Suppose $\phi_{t_i} \in \mathcal{F}_{t_i}$ and $\mathbb{E}[\phi_{t_i}^2] < \infty$. Prove that

$$\mathbb{E}[\phi_{t_i} \Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

(d) Explain why the last identity prevents a predictable Brownian trading strategy from creating positive expected gain, but does not prevent it from changing risk.

Exercise 3.2 (Level 1 — Deterministic and random step strategies)**.** Consider the processes

$$\phi_t = 2\mathbf{1}_{(0,1]}(t) - \mathbf{1}_{(1,3]}(t)$$

and

$$\psi_t = \mathbf{1}_{\{W_1>0\}}\mathbf{1}_{(1,2]}(t).$$

(a) Verify that both processes are predictable.

(b) Write $\int_0^3 \phi_t \, \mathrm{d}W_t$ and $\int_0^2 \psi_t \, \mathrm{d}W_t$ without integral notation.

(c) Identify the law of the first integral.

(d) Compute the mean and second moment of the second integral.

(e) Show that the second integral has an atom at zero and identify its law as a mixture. Why is it not Gaussian?

(f) Explain why deterministic integrands behave differently from random predictable integrands with respect to Gaussianity.

Part II — Isometry and the L^2 Construction

Exercise 3.3 (Level 2 — Prove the Itô isometry)**.** Let

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i,t_{i+1}]}(t), \qquad \xi_i \in \mathcal{F}_{t_i}, \qquad \mathbb{E}[\xi_i^2] < \infty.$$

Set $I(\phi) = \sum_i \xi_i \Delta W_i$.

(a) Prove that $\mathbb{E}[I(\phi)] = 0$.

(b) If $i < j$, prove carefully that

$$\mathbb{E}[\xi_i \Delta W_i \, \xi_j \Delta W_j] = 0$$

by conditioning on $\mathcal{F}_{t_j}$.

(c) Prove the isometry

$$\mathbb{E}[I(\phi)^2] = \mathbb{E}\int_0^T \phi_t^2 \, \mathrm{d}t.$$

(d) For another simple predictable process ψ, derive

$$\mathbb{E}[I(\phi)I(\psi)] = \mathbb{E}\int_0^T \phi_t \psi_t \, \mathrm{d}t.$$

(e) If $|\phi_t| \le L$, prove the risk bound

$$\mathrm{Var}(I(\phi)) \le L^2 T.$$

Exercise 3.4 (Level 2 — Approximation and an exact L^2 error)**.** Let $f(t) = t$ on $[0,T]$, take the uniform grid $t_i = iT/n$, and define the left-step approximation

$$f_n(t) = \sum_{i=0}^{n-1} t_i \mathbf{1}_{(t_i,t_{i+1}]}(t).$$

(a) Compute

$$\int_0^T |f(t) - f_n(t)|^2 \, \mathrm{d}t$$

exactly.

(b) Deduce from the isometry that

$$\sum_{i=0}^{n-1} t_i \Delta W_i \longrightarrow \int_0^T t \, \mathrm{d}W_t \quad \text{in } L^2(\Omega).$$

Give the exact root-mean-square error.

(c) Explain why the isometry makes the integral independent of the chosen sequence of simple predictable approximations.

(d) On $[0, 1]$, compare

$$\phi_t = 1, \qquad \psi_t = \sqrt{2}\, \mathbf{1}_{(0,1/2]}(t).$$

Show that the two terminal integrals have the same law but that their gain processes do not accumulate risk at the same times.

Part III — Martingales, Integrability, and Stopping

Exercise 3.5 (Level 2 — Audit the martingale assumptions)**.** Let ϕ be predictable and suppose

$$\mathbb{E} \int_0^T \phi_s^2 \, \mathrm{d}s < \infty.$$

Define $M_t = \int_0^t \phi_s \, \mathrm{d}W_s$.

(a) Use the isometry and Cauchy–Schwarz to prove that

$$\mathbb{E}|M_t| \le \left(\mathbb{E} \int_0^t \phi_s^2 \, \mathrm{d}s \right)^{1/2} < \infty.$$

Explain why checking $M_t \in L^1$ is logically necessary before using a conditional expectation in the definition of a martingale.

(b) Prove the martingale identity first for a simple predictable process, then explain how L^2 approximation allows passage to a general ϕ.

(c) Show that L^2 convergence implies the L^1 convergence needed to pass conditional expectations to the limit.

(d) State clearly the difference between the assumptions

$$\mathbb{E} \int_0^T \phi_s^2 \, \mathrm{d}s < \infty \qquad \text{and} \qquad \int_0^T \phi_s^2 \, \mathrm{d}s < \infty \quad \text{a.s.}$$

What type of martingale is guaranteed by each condition?

Exercise 3.6 (Level 3 — Pathwise finite energy need not give a true martingale)**.** Let $X \ge 0$ be finite almost surely, independent of W, and satisfy $\mathbb{E}[X] = \infty$. Enlarge the filtration so that $X \in \mathcal{F}_0$, and set $\phi_t = X$.

(a) Show that ϕ is predictable and

$$\int_0^T \phi_t^2 \, \mathrm{d}t = X^2 T < \infty \quad \text{a.s.}$$

(b) Show formally, and then through localization, that

$$M_t = \int_0^t X \, \mathrm{d}W_s = X W_t.$$

(c) Prove that $\mathbb{E}|M_t| = \infty$ for every $t > 0$. Deduce that M is not a true martingale.

(d) For $n \geq 1$, define

$$\tau_n = \begin{cases} \infty, & X \leq n, \\ 0, & X > n. \end{cases}$$

Verify that $\tau_n \uparrow \infty$ almost surely and that $(M_{t\wedge\tau_n})_{t\geq 0}$ is a square-integrable martingale. Conclude that M is a local martingale.

(e) Explain how the conclusions change if $\mathbb{E}[X^2] < \infty$.

Exercise 3.7 (Level 2 — Conditional risk of a state-dependent strategy)**.** On $[0, 2]$, define

$$\phi_t = \mathbf{1}_{(0,1]}(t) + \left(1 + \mathbf{1}_{\{W_1 > 0\}}\right)\mathbf{1}_{(1,2]}(t), \qquad M_t = \int_0^t \phi_s \, \mathrm{d}W_s.$$

(a) Verify predictability and expected finite energy.

(b) Compute M_1 and write $M_2 - M_1$ explicitly.

(c) Compute $\mathbb{E}[M_2 \mid \mathcal{F}_1]$.

(d) Find $\mathrm{Var}(M_2 - M_1 \mid \mathcal{F}_1)$ and the conditional law of M_2 given $\mathcal{F}_1$.

(e) Compute $[M]_2$ and interpret it as a state-dependent exposure clock.

Exercise 3.8 (Level 2 — Stop the exposure at a barrier)**.** Let τ be a stopping time and define

$$M_t = \int_0^t \mathbf{1}_{\{s \leq \tau\}} \, \mathrm{d}W_s.$$

(a) Give the trading interpretation of the integrand and show that $M_t = W_{t\wedge\tau}$.

(b) For fixed finite t, verify

$$\mathbb{E} \int_0^t \mathbf{1}_{\{s\leq\tau\}}^2 \, \mathrm{d}s = \mathbb{E}[t \wedge \tau] \leq t.$$

Explain why this is the relevant integrability check.

(c) Deduce

$$\mathbb{E}[W_{t\wedge\tau}] = 0, \qquad \mathbb{E}[W_{t\wedge\tau}^2] = \mathbb{E}[t \wedge \tau].$$

(d) Apply the result to $\tau_a = \inf\{u \geq 0 : W_u = a\}$, with $a > 0$, without using the density of τ_a.

(e) Explain why these conclusions do not require τ itself to be bounded.

Part IV — Deterministic Gaussian Applications

Exercise 3.9 (Level 2 — Gaussian projection and independent risks)**.** Let

$$I_T = \int_0^T t\,\mathrm{d}W_t.$$

(a) Identify the law of I_T.

(b) Write W_T as a stochastic integral and compute $\mathrm{Cov}(I_T, W_T)$.

(c) Find $\mathbb{E}[I_T \mid W_T]$ and $\mathrm{Var}(I_T \mid W_T)$.

(d) Find an affine function $g(t) = t - c$ such that

$$Y = \int_0^T g(t)\,\mathrm{d}W_t$$

is independent of W_T, and compute $\mathrm{Var}(Y)$.

(e) Interpret the preceding calculations as orthogonal projection in $L^2(0,T)$.

(f) For $M_t = \int_0^t s\,\mathrm{d}W_s$, show that

$$(M_t)_{t\geq 0} \overset{\text{law}}{=} (B_{t^3/3})_{t\geq 0}$$

for a Brownian motion B.

Exercise 3.10 (Level 2 — Deterministic integration by parts and Brownian area)**.** Let $f : [0,T] \to \mathbb{R}$ be deterministic and absolutely continuous with $f' \in L^2(0,T)$.

(a) Explain why $f' \in L^1(0,T)$, why f has finite variation, and why $f \in L^2(0,T)$.

(b) Starting from discrete summation by parts, prove

$$f(T)W_T - f(0)W_0 = \int_0^T f(t)\,\mathrm{d}W_t + \int_0^T W_t f'(t)\,\mathrm{d}t.$$

Make clear which limit is in L^2 and which is pathwise.

(c) Explain why no general Itô formula and no quadratic-variation correction are needed.

(d) With $f(t) = t$, prove

$$\int_0^T W_t\,\mathrm{d}t = TW_T - \int_0^T t\,\mathrm{d}W_t.$$

(e) Identify the law of the Brownian area $A_T = \int_0^T W_t\,\mathrm{d}t$.

Part V — Conventions and Simulation

Exercise 3.11 (Level 2 — Left, right, and midpoint sums)**.** For a partition π of $[0,T]$, define

$$L^\pi = \sum_i W_{t_i}\Delta W_i, \quad R^\pi = \sum_i W_{t_{i+1}}\Delta W_i, \quad S^\pi = \sum_i \frac{W_{t_i} + W_{t_{i+1}}}{2}\Delta W_i.$$

(a) Use

$$W_{t_{i+1}}^2 - W_{t_i}^2 = 2W_{t_i}\Delta W_i + (\Delta W_i)^2$$

to identify the limit of L^π.

(b) Deduce the limits of R^π and S^π.

(c) Compute the mean of each limit. Which convention preserves the zero-mean martingale-gain interpretation?

(d) Let $I_T = \int_0^T W_t \, dW_t$. Compute $\mathbb{E}[I_T^2]$ from the explicit formula and again from the isometry.

(e) Explain exactly what is illegal about choosing $\phi_{t_i} = \operatorname{sign}(\Delta W_i)$.

Exercise 3.12 (Level 2 — Simulation error and implementation checks)**.** On the uniform grid $t_i = iT/n$, simulate

$$\Delta W_i = \sqrt{\Delta t}\, Z_i, \qquad Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad \Delta t = T/n.$$

Define

$$I^{(n)} = \sum_{i=0}^{n-1} W_{t_i} \Delta W_i, \qquad I = \int_0^T W_t \, dW_t.$$

(a) Prove the exact error identity

$$I - I^{(n)} = \frac{1}{2} \sum_{i=0}^{n-1} \big((\Delta W_i)^2 - \Delta t\big).$$

(b) Derive

$$\mathbb{E}[(I - I^{(n)})^2] = \frac{T^2}{2n} \qquad \text{and} \qquad \operatorname{RMSE}(I^{(n)}) = \frac{T}{\sqrt{2n}}.$$

(c) For a state-dependent integrand $\phi_t = h(W_t)$, write the correct order in which ϕ_i, I_{i+1}, and $W_{t_{i+1}}$ must be updated.

(d) Explain the effect of each implementation error:

(i) using $\Delta t\, Z_i$ instead of $\sqrt{\Delta t}\, Z_i$;

(ii) computing ϕ_i from $W_{t_{i+1}}$;

(iii) using independent Brownian increments for the state and its stochastic integral;

(iv) replacing ΔW_i by Δt.

(e) Propose at least four diagnostics involving means, second moments, covariance, grid refinement, or exact benchmarks. Explain why inspecting one plausible path is not sufficient.